



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal TO KRC FUELS

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 152.81 km**  
**Estimated Duration: 2.7 hours**  
**Adjusted Duration (Heavy Vehicle): 3.4 hours**  
**Start: (11.4381, 77.8734)**  
**End: (12.521272, 78.228036)**



# Welcome to the Journey Risk Management Study

## 1. Overview of the Route Map

The route from Sangagiri to Krishnagiri spans approximately 153 kilometers, heading east-northeast. It primarily follows the NH44 highway, a significant transport artery extending through Tamil Nadu. The journey passes through semi-urban and rural areas, with sections of the route transitioning from major highways to more regional roads as you approach Krishnagiri.



## 2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences tropical weather conditions. Along this route:

- **Summer (March to June):** High temperatures can lead to roadway heat haze.
- **Monsoon (July to September):** Expect heavy rain, which may cause flooding, especially in low-lying areas.
- **Winter (November to February):** Mild weather with the potential for fog in the early mornings. Potential hazards include reduced visibility during rains and slippery road surfaces.

## 3. Analysis of Traffic Patterns

- **Peak Hours:** Mornings (7:00-10:00 AM), and evenings (6:00-9:00 PM), especially near urban centers like Salem.
- **Congestion-Prone Areas:** Outskirts of Salem and other small towns. Roadworks and local traffic can cause delays.

## 4. Assessment of Road Quality and Infrastructure

- **NH44 is well-maintained**, with good signage and ample service facilities.
- Some sections may have narrower lanes and less lighting, predominantly when entering smaller towns.
- Regular checks for potholes and surface wear are needed post-monsoon.

## 5. Suggestions for Alternative Routes for Emergencies

- Consider using the NH544 towards Salem and transition back onto alternate regional roads leading into Krishnagiri as needed.
- Local roads near Dharmapuri can provide further diversion options, particularly in heavy traffic or roadblock scenarios.

## 6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Permits Required:** Ensure all appropriate permits for transporting hazardous materials are obtained.
- **Timing Restrictions:** Transport of hazardous materials may be restricted during peak hours or specific local regulations.

## 7. Overview of Historical Incidents

- Occasional incidents involving rural vehicles and heavy trucks near urban centers.
- Few documented incidents of hazardous material spills due to enforcement of strict transport regulations.

## 8. Environmental Considerations and Sensitive Areas

- **Agricultural Regions:** Be mindful of tractor crossings and farm animals.
- **Wildlife Areas:** Few, but be alert in peripheral zones where wildlife may cross at night.

## 9. Analysis of Communication Coverage

- Major towns and highways have good cellular coverage.
- Potential dead zones: Rural stretches away from major towns or elevated terrains.

## 10. Estimated Emergency Response Times

- **Urban Areas (e.g., Salem):** Approximately 20-30 minutes.
- **Rural Areas:** Up to 60 minutes, depending on proximity to the nearest town.

## 12. Overall Summary of Risk Assessment

- **Route Safety:** Generally safe with manageable risks if driving conditions and regulations are respected.
- **Main Concerns:** Weather-induced risks, traffic congestion in urban areas, and occasional infrastructure wear.
- **Recommendations:** Regular updates on weather forecasts and traffic, and being equipped with emergency contact numbers for local services and communication tools for dead zones.

Overall, this route is relatively secure for transporting hazardous materials when well-prepared, taking into account the noted variables and adhering to local laws and regulations.

### Risk Assessment - Turns

|   | Risk Type | Risk Level | Coordinates        | Speed Limit |
|---|-----------|------------|--------------------|-------------|
| 0 | Turn      | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 1 | Turn      | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 2 | Turn      | High       | 11.44900, 77.87406 | 15 KM/Hr    |
| 3 | Turn      | High       | 11.45028, 77.87202 | 15 KM/Hr    |
| 4 | Turn      | Medium     | 11.46400, 77.87323 | 30 KM/Hr    |
| 5 | Turn      | High       | 11.47403, 77.86841 | 15 KM/Hr    |
| 6 | Turn      | High       | 11.49149, 77.89601 | 15 KM/Hr    |
| 7 | Turn      | High       | 11.49224, 77.89601 | 15 KM/Hr    |
| 8 | Turn      | Medium     | 11.53501, 77.95707 | 30 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 9  | Turn       | Medium     | 11.53601, 77.95856 | 30 KM/Hr    |
| 10 | Turn       | Medium     | 11.62403, 78.10893 | 30 KM/Hr    |
| 11 | Turn       | High       | 11.64180, 78.11808 | 15 KM/Hr    |
| 12 | Turn       | High       | 11.64325, 78.11993 | 15 KM/Hr    |
| 13 | Turn       | Medium     | 11.69519, 78.10775 | 30 KM/Hr    |
| 14 | Turn       | Medium     | 11.73958, 78.05105 | 30 KM/Hr    |
| 15 | Turn       | Medium     | 11.75835, 78.05016 | 30 KM/Hr    |
| 16 | Turn       | Medium     | 11.86847, 78.08367 | 30 KM/Hr    |
| 17 | Turn       | Medium     | 11.87942, 78.08197 | 30 KM/Hr    |
| 18 | Turn       | Medium     | 11.92585, 78.06507 | 30 KM/Hr    |
| 19 | Turn       | Medium     | 11.93704, 78.05402 | 30 KM/Hr    |
| 20 | Turn       | High       | 11.95818, 78.04978 | 15 KM/Hr    |
| 21 | Turn       | Medium     | 11.95988, 78.04948 | 30 KM/Hr    |
| 22 | Turn       | Medium     | 11.96304, 78.05136 | 30 KM/Hr    |
| 23 | Turn       | High       | 11.96399, 78.05135 | 15 KM/Hr    |
| 24 | Turn       | High       | 11.96580, 78.04923 | 15 KM/Hr    |
| 25 | Turn       | Medium     | 11.96675, 78.04939 | 30 KM/Hr    |
| 26 | Turn       | Medium     | 11.96791, 78.05070 | 30 KM/Hr    |
| 27 | Turn       | Medium     | 12.19881, 78.18157 | 30 KM/Hr    |
| 28 | Turn       | Medium     | 12.22459, 78.19136 | 30 KM/Hr    |
| 29 | Turn       | Medium     | 12.25060, 78.19977 | 30 KM/Hr    |
| 30 | Turn       | Medium     | 12.25329, 78.20039 | 30 KM/Hr    |
| 31 | Turn       | Medium     | 12.29110, 78.19909 | 30 KM/Hr    |
| 32 | Turn       | Medium     | 12.40589, 78.22275 | 30 KM/Hr    |
| 33 | Turn       | High       | 12.50709, 78.22230 | 15 KM/Hr    |
| 34 | Blind Spot | Blind Spot | 12.52371, 78.23155 | 10 KM/Hr    |

Emergency Locations

|   | type     | name              | coordinates            | speed_limit | risk_level |
|---|----------|-------------------|------------------------|-------------|------------|
| 0 | hospital | Thilagam Hospital | 11.4753573, 77.8685829 | 30 km/h     | Medium     |

|    | type     | name                                                            | coordinates               | speed_limit | risk_level |
|----|----------|-----------------------------------------------------------------|---------------------------|-------------|------------|
| 1  | clinic   | Meyyazhagan Clinic                                              | 11.4752254,<br>77.8685671 | 30 km/h     | Medium     |
| 2  | hospital | Dr. Jaganathan Hospital                                         | 11.4752717,<br>77.8709312 | 30 km/h     | Medium     |
| 3  | hospital | Kathir Eye Hospital                                             | 11.4740976,<br>77.8681591 | 30 km/h     | Medium     |
| 4  | police   | Police Quarters                                                 | 11.4741243,<br>77.8673495 | 30 km/h     | Medium     |
| 5  | hospital | Sankari Hospital                                                | 11.4777496,<br>77.873892  | 30 km/h     | Medium     |
| 6  | clinic   | Dr. Dhanasekar Clinic                                           | 11.4793113,<br>77.8829247 | 30 km/h     | Medium     |
| 7  | hospital | Sri Vellaiyan Hospital                                          | 11.4798876,<br>77.8831394 | 30 km/h     | Medium     |
| 9  | hospital | C. N. Hospital                                                  | 11.55724, 78.0096         | 30 km/h     | Medium     |
| 10 | hospital | Vinayaka Mission's Homeopathic Medical College & Hospital       | 11.583617,<br>78.059145   | 30 km/h     | Medium     |
| 11 | hospital | Vinayaka mission Kirubananadah variyar medical college hospital | 11.584984,<br>78.0618417  | 30 km/h     | Medium     |
| 12 | hospital | vims                                                            | 11.5854239,<br>78.0639188 | 30 km/h     | Medium     |
| 14 | hospital | Vinayaka Mission’s Kirupananda Variyar Medical College          | 11.5977669,<br>78.0850992 | 30 km/h     | Medium     |
| 19 | hospital | Bharathi Hospital, Salem                                        | 11.6492647,<br>78.1204131 | 30 km/h     | Medium     |
| 20 | hospital | Sisu Hospital Inida Ltd                                         | 11.6505276,<br>78.1205003 | 30 km/h     | Medium     |
| 21 | hospital | Aishwarya Hospital                                              | 11.6716291,<br>78.1267807 | 30 km/h     | Medium     |
| 22 | hospital | C.M. Hospital                                                   | 11.6747345,<br>78.1255034 | 30 km/h     | Medium     |
| 24 | pharmacy | Healthy Pharmacy                                                | 11.6762714,<br>78.127744  | 30 km/h     | Medium     |
| 25 | pharmacy | Thulasi Pharmacy                                                | 11.6761991,<br>78.1277377 | 30 km/h     | Medium     |
| 26 | hospital | Sri Gokulam Speciality Hospital                                 | 11.6788724,<br>78.1274258 | 30 km/h     | Medium     |
| 27 | hospital | Global Medical Center                                           | 11.6867049,<br>78.1186697 | 30 km/h     | Medium     |

|    | type     | name                               | coordinates               | speed_limit | risk_level |
|----|----------|------------------------------------|---------------------------|-------------|------------|
| 28 | hospital | Manipal Hospital                   | 11.703072,<br>78.1026612  | 30 km/h     | Medium     |
| 31 | clinic   | Manimuthu Skin Clinic              | 11.7521189,<br>78.0484058 | 30 km/h     | Medium     |
| 32 | hospital | St. Joseph's Speciality Hospital   | 11.761611,<br>78.0522344  | 30 km/h     | Medium     |
| 33 | hospital | Anugraha Multi Speciality Hospital | 11.773549,<br>78.0546408  | 30 km/h     | Medium     |
| 34 | hospital | L.R. Hospital                      | 11.8703354,<br>78.0844554 | 30 km/h     | Medium     |
| 35 | clinic   | JSA Clinic                         | 11.8700619,<br>78.0844982 | 30 km/h     | Medium     |
| 37 | hospital | Government Hospital                | 12.3048002,<br>78.2060881 | 30 km/h     | Medium     |
| 42 | hospital | TCR Multi Speciality Hospital      | 12.5088183,<br>78.2237657 | 30 km/h     | Medium     |

## Crowded Spots

|    | type       | name                                                    | coordinates               | speed_limit | risk_level |
|----|------------|---------------------------------------------------------|---------------------------|-------------|------------|
| 13 | school     | Vinayaka Missions Annapoorani Matric School             | 11.5971662,<br>78.0839876 | 30 km/h     | Medium     |
| 15 | university | Vinayaka Missions Vinayaka Missions Research Foundation | 11.5974056,<br>78.0843617 | 30 km/h     | Medium     |
| 16 | school     | Dist Education Training                                 | 11.6133799,<br>78.1022409 | 30 km/h     | Medium     |
| 17 | school     | DIET - Salem                                            | 11.6144158,<br>78.1025078 | 30 km/h     | Medium     |
| 18 | school     | Diamond Rays Matric HR Sec School                       | 11.6385629,<br>78.1206259 | 30 km/h     | Medium     |
| 23 | college    | Sona College                                            | 11.6768,<br>78.1257052    | 30 km/h     | Medium     |
| 29 | university | Periyar University                                      | 11.7186758,<br>78.0771089 | 30 km/h     | Medium     |
| 30 | school     | Fathima Hr Sec School                                   | 11.7410508,<br>78.0479272 | 30 km/h     | Medium     |



|    | type    | name                                    | coordinates               | speed_limit | risk_level |
|----|---------|-----------------------------------------|---------------------------|-------------|------------|
| 36 | college | Laxminarayana Polytechnic College       | 12.0421277,<br>78.103203  | 30 km/h     | Medium     |
| 38 | school  | Government Boys Higher secondary school | 12.3047059,<br>78.2065495 | 30 km/h     | Medium     |
| 39 | school  | Nalanda Metric Higher Secondary School  | 12.488407,<br>78.21856    | 30 km/h     | Medium     |
| 40 | school  | Nalanda International Public School     | 12.4994979,<br>78.2235094 | 30 km/h     | Medium     |
| 41 | school  | Don Bosco School Ground                 | 12.505954,<br>78.2222003  | 30 km/h     | Medium     |

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.44900, 77.87406



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45028, 77.87202



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.46400, 77.87323





Risk Type: Turn  
Risk Level: High  
Speed Limit: 15 KM/Hr  
Coordinates: 11.47403, 77.86841



Risk Type: Turn  
Risk Level: High  
Speed Limit: 15 KM/Hr  
Coordinates: 11.49149, 77.89601





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.49224, 77.89601



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.53501, 77.95707



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.53601, 77.95856





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.62403, 78.10893



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.64180, 78.11808





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.64325, 78.11993



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.69519, 78.10775



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.73958, 78.05105





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.75835, 78.05016



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.86847, 78.08367





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.87942, 78.08197



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.92585, 78.06507



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.93704, 78.05402





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.95818, 78.04978



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.95988, 78.04948





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96304, 78.05136



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.96399, 78.05135



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.96580, 78.04923





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96675, 78.04939



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96791, 78.05070





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.19881, 78.18157



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.22459, 78.19136



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.25060, 78.19977





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.25329, 78.20039



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.29110, 78.19909





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.40589, 78.22275



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.50709, 78.22230



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 12.52371, 78.23155



